

Pritish Chakraborty

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EDUCATION

- **Indian Institute of Technology, Bombay** Mumbai, Maharashtra
MS + PhD (dual degree) in Computer Science; GPA: 9.14/10 July 2021 – June 2024 — July 2024 - present
- **Manav Rachna International University** Delhi NCR
BTech in Computer Science and Engineering; GPA: 8.27/10 July 2010 – May 2014

RESEARCH INTERESTS

Information Retrieval for Structured Objects, Graph Retrieval and interfacing with LLMs, ML on Graphs, Neural Tools for Combinatorial Algorithms, Geometric Deep Learning, Generative Models, Applications (AI4Science)

RESEARCH WORK

- **A Dense Subset Index for Collective Query Coverage (DISCo)** June 2025 - October 2025.
Advisors: Prof. Abir De and Prof. Soumen Chakrabarti
Developed dense retrieval framework for **collective query coverage** that retrieves a set of documents jointly, so that different aspects of a query are covered, as opposed to independent ranking. Includes contributions to the core indexing and retrieval mechanism, various definitive baselines and ablation experiments, and performance benchmarking of the end-to-end system. **Paper accepted to ICLR 2026.** (Please see: <https://tinyurl.com/disco-paper>)
- **Contextual Tokenization for Graph Inverted Indices (CoRGII)** February 2025 - May 2025
Advisors: Prof. Abir De and Prof. Soumen Chakrabarti
Developed **efficient, differentiable sparse retrieval pipeline** for the purpose of graph retrieval. Consists of neural tokenizer for discretized latent vocabulary, impact scorer network for differentiable analog of TF-IDF scoring, and multi-probe expansion using GPU-bound inverted index for greater recall. Leverages contextual dense graph representations to generate discrete codes, which support highly optimized inverted indices. **Paper accepted to NeurIPS 2025.** (Please see: <https://pritishc.com/posts/2025/10/corgii-blog/>)
- **Differentiable Adversarial Attacks against Marked Temporal Point Processes (PermTPP)** Nov 2023 - June 2024
Advisors: Prof. Abir De and Prof. Srikanth Bedathur
Developed new class of **fully differentiable** adversarial attacks for marked temporal point process (MTPP) models, while outlining why standard attacks (PGD, etc) are not suitable. The attack corrupts input continuous-time event sequence data in a way that perturbs order relationship between elements of a sequence. Subsequently, developed algorithm to train MTPP models to become robust to this class of attacks. Evaluated the method on four real-world datasets, in addition to datasets from specific domains. Results show that our method outperforms standard attacks. **Paper accepted to AAAI 2025.** (Please see: <https://tinyurl.com/mtpp-adv>)
- **Learning and Maximizing Influence in Social Networks Under Capacity Constraints** Jan 2022 - Aug 2022.
Advisors: Prof. Abir De and Prof. Sayan Ranu
Developed and implemented an MC simulation based method to obtain an (approximately) optimal solution to the **top-K influence maximization problem**. Helped develop **alpha-submodular** theoretical guarantees in the process. Developed **differentiable deep submodular neural network** to predict the top-K maximized influential nodes in a graph. Both simulator and ML models outperformed state-of-the-art influence maximization methods. **Paper accepted to WSDM 2023.** (Please see: <https://tinyurl.com/topk-im>)

TECHNICAL SKILLS

- **Languages:** Python, Javascript, C/C++, SQL, other languages such as C# and Java
- **Technologies:** Pytorch, Pytorch Geometric, Django, Angular.js, React, REST, other web technologies

COURSES TAKEN

Foundations of Machine Learning, Algorithms and Complexity, Retrieval on Text and Graphs, Advanced Machine Learning, Optimization in Machine Learning, Organization of Web Information (NLP+IR), Real Analysis, Basic Algebra

WORK EXPERIENCE

- **Google Research India** Hybrid
Student Researcher *July 2023 - November 2023*
 - **Responsibilities** End-to-end research and execution. Building Google-specific tools based on my work on influence maximization.
 - **Projects** Research on the robust influence maximization problem. Development of general foundation models for information diffusion. Development of fast neural approximators for replacement of MC simulations.
- **Instahyre** New Delhi, India
Software Engineering/Head of Engineering *July 2015 - May 2019*
 - **Responsibilities** Handled the complete tech stack and deployment pipeline for multiple feature release cycles. Mentored and brought junior engineers upto speed.
 - **Research and Execution** Undertook systems research for various aspects of the codebase, both during maintenance and when extending features from scratch. Aligned this research with business agenda - for example, deciding to scale parts of the system, vertically or horizontally, depending on projected user count and user load across time. Additionally, undertook user experience research when deciding to release certain big-ticket features, for example performing large scale A/B tests.
 - **Projects** Worked on several important features - the matching engine, the Elasticsearch-powered search platform, internal tools - end to end. Took responsibility for a streamlined UX and production management.

EXTRA-CURRICULARS

- Won the **ACM IARCS Research Travel Grant** for the NeurIPS 2025 paper.
- Won the **NeurIPS 2025 Scholar Award**, which covers registration and accommodation during the conference.
- Won the **Qualcomm Fellowship** in the 2025 cycle.
- Selected to attend Machine Learning Summer Schools (mlss.cc) and London Geometry and Machine Learning Summer School (logml.ai), both highly competitive. Our group won the **prize for best project** at LogML.
- Attended selective **Microsoft Research India Summit** on ML research in Bengaluru (*July'25*).
- Attended selective three-day **Google Research Week** symposium on ML research in Bengaluru (*Jan'23, Feb'24, Feb'25*).